

PROFILING CYBER-VICTIMS

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Outline

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2. From victimology to cyber-victimology
3. Measuring cyber-crime through the victims' lens
4. Explaining cyber-crime victimisation
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1. What is victimology?

What is victimology?

- term used for the first time in 1947 by B. Mendelsohn to describe the **scientific study of crime victims**
- subfield of criminology: study of causes/etiology of victimisation, its consequences, how criminal justice systems/other actors deal with victims
- initially, **focus on the role of the victim**: victim precipitation/facilitation/provocation
- 1960s: **more sympathetic view** of crime victims, thanks to the victims rights' movements
- contributions of the victims rights' movements
- victimology today

2. From victimology to cyber-victimology

The unique traits of cybervictimisation

Cybercrime victimisation has unique traits:

- defies the logic of physical proximity
- automation makes it possible to victimise several persons with the same act

“Human” element and harm of cybercrime was recognised only in 2001, by the CoE Convention on Cyber Crime

Typology of Cyber Victimization

Jaishankar (2012, 2013) proposed the following classification of cybercrimes:

1. Cybervictimisation of governments
2. Cybervictimisation of corporations
3. Cybervictimisation of individuals or interpersonal cyber victimisation
4. Victimless crimes

3. Measuring cyber-crime through the victims' lens

Victimisation surveys and cybercrime

- What about cybercrime victimisation surveys?

Reep-van den Bergh and Junger *Crime Sci* (2018) 7:5
<https://doi.org/10.1186/s40163-018-0079-3>

 Crime Science

RESEARCH

Open Access



Victims of cybercrime in Europe: a review of victim surveys

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Victimisation surveys and cybercrime

- Objective: to map existing cyber-victimization surveys in Europe, to provide estimates of annual prevalence rates (n. of victims per population) for 6 cyber-crimes
- Methodology: definition of inclusion criteria for surveys; Search in databases and on the Internet, grey literature, contacts with national statistical offices
- Results: 9 surveys identified; different annual prevalence rates depending on the crime: e.g. 1-3% for online shopping fraud; 3% of the population is a victim of cyberbullying

3. Explaining cyber-crime victimisation

Theories of cyber-crime victimisation

1. Lifestyle theory (Hindelang et al., 1978): certain behaviours (related to work, school and leisure) expose individuals to risky (crime-prone) situations - **certain behaviors elevate one's odds of being victimized**
2. Routine activity theory (RAT) (Cohen & Felson, 1979): victimization results from convergence in time and space of a motivated offender, a suitable target/victim and the absence of capable guardianship (humans or objects that can prevent crime from occurring) – **describes victimisation**

Although 1. and 2. are distinct, they lead to similar hypotheses and are often combined (lifestyle routine activity theory – LRAT)

Theories of cybervictimisation

- Reyns et al. (2011) revisited LRAT to adapt it to cybercrime; this resulted in the **cyberlifestyle-routine activity theory**:
 - ✓ convergence in place: criminal and victims are often physically separate, still they come together in the same virtual space, via computer networks
 - ✓ convergence in time: offender and victim may not interact in real time, still they intersect in time (e.g. the offender sends an email that the victim reads some time later)

Theories of cyber-crime victimisation

3. General theory of crime (Gottfredson, Hirschi, 1990): key factor in causing crime and deviance is **low self-control**, i.e. the inability by an individual to exercise personal restraint in the face of tempting immediate and easy gratification both in the short and long-term – **developed for offending, adapted to victimisation by Schreck in 1999**
4. Actor-Network Theory (van der Wagen, Pieters, 2018): **to account for emerging forms of high-tech cybercrime** (such as ransomware, botnets and virtual theft) in which the **offender targets a hybrid victim (composite of human, technical and virtual entities)**

5. Who are cyber-victims? Findings from scientific literature

Many studies, many research questions and methods

- different studies, **different research questions and methods** to investigate cyber-victims (e.g. individuals vs business victimisation; focus on different cybercrime types e.g. cyberbullying, identity theft; focus group vs survey/interviews)
- overall, limited amount of studies
- sometimes conflicting findings

A routine activities analysis

- study by Junger et al. (2017)
- aim: to investigate relationship between users' routine activities and socio-economic characteristics and three forms of cybercrime victimisation (online shopping fraud, online banking fraud, DDoS)
- methodology: data from Eurobarometer (sample of 17,811 European citizens); logistic regression
- results: 1) generally support RAT; 2) show that socio-economic variables (gender, age, economic factors) are almost completely irrelevant: **normalization of cybervictimization**

6. Focusing on repeat victimisation

Focusing on repeat victimisation

- Repeat victimization = experiencing more than one victimization
- “People, places, products or services, vehicle, household, business, or systems (including virtual networks) are, once victimized, more likely to be victimized again. Further victimization by the same type of crime is most likely” (Pease, 2018)
- **Supertargets:** 1 or 2 % of Potential Targets experience around half of crime

PREDICTING CRIME

Many studies show that previous victimization is an risk indicator.

In particular, there are various studies conducted in the United Kingdom on residential burglaries.

Key question: "Do residential burglaries cluster in time and space?"

Theoretical basis

- optimal foraging theory: animals optimize foraging strategies to maximize benefits while minimizing risks and search time
- transmissibility (from epidemiology) - closeness in space and time of epidemics/diseases...

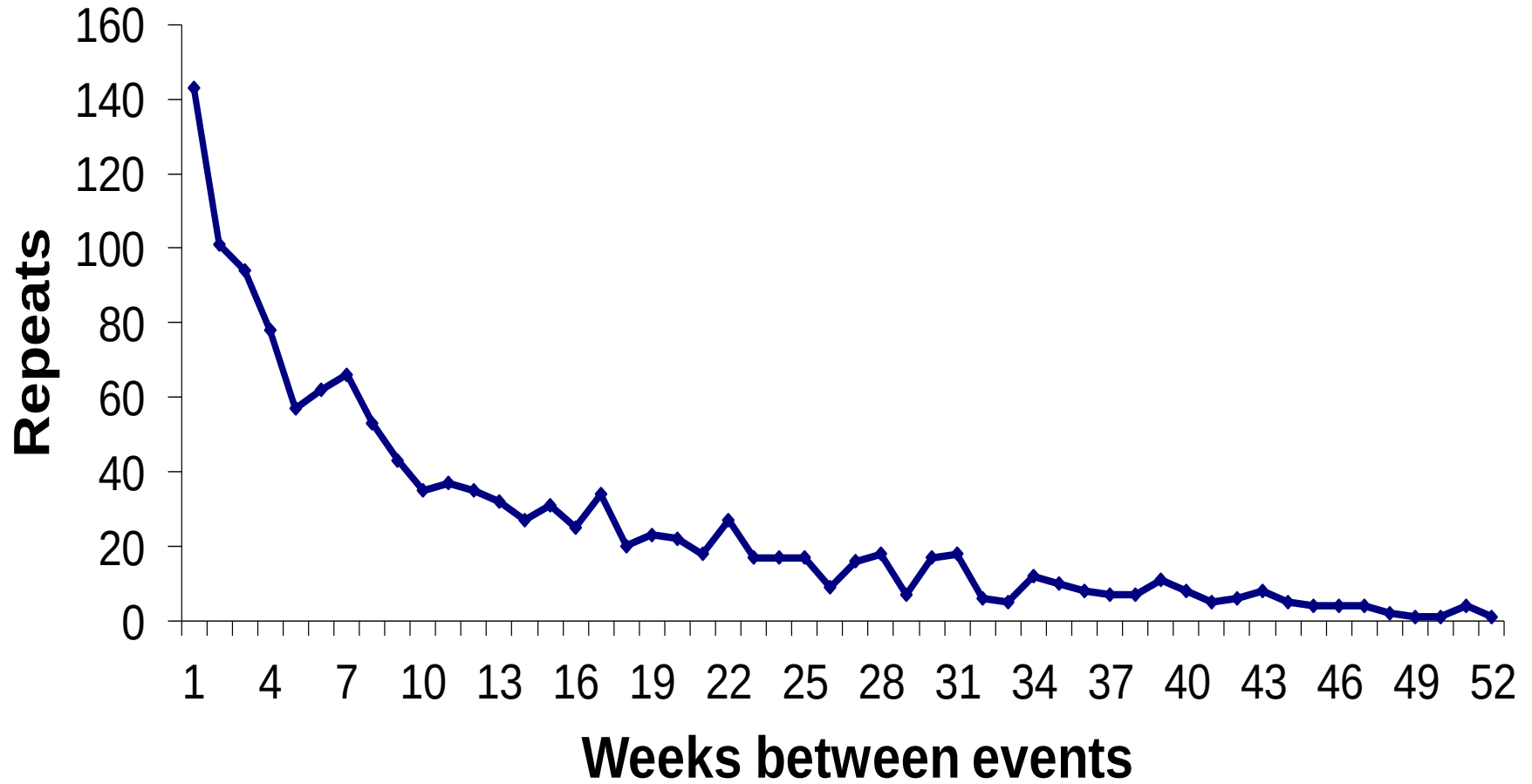
does this also apply to home burglary?

Methodology

- georeferenced data on burglaries in Merseyside (6 years)
- statistical tests to verify whether the space-time clustering was a random event or not
- for each street, calculation of the space-time distance between the break-ins

Results

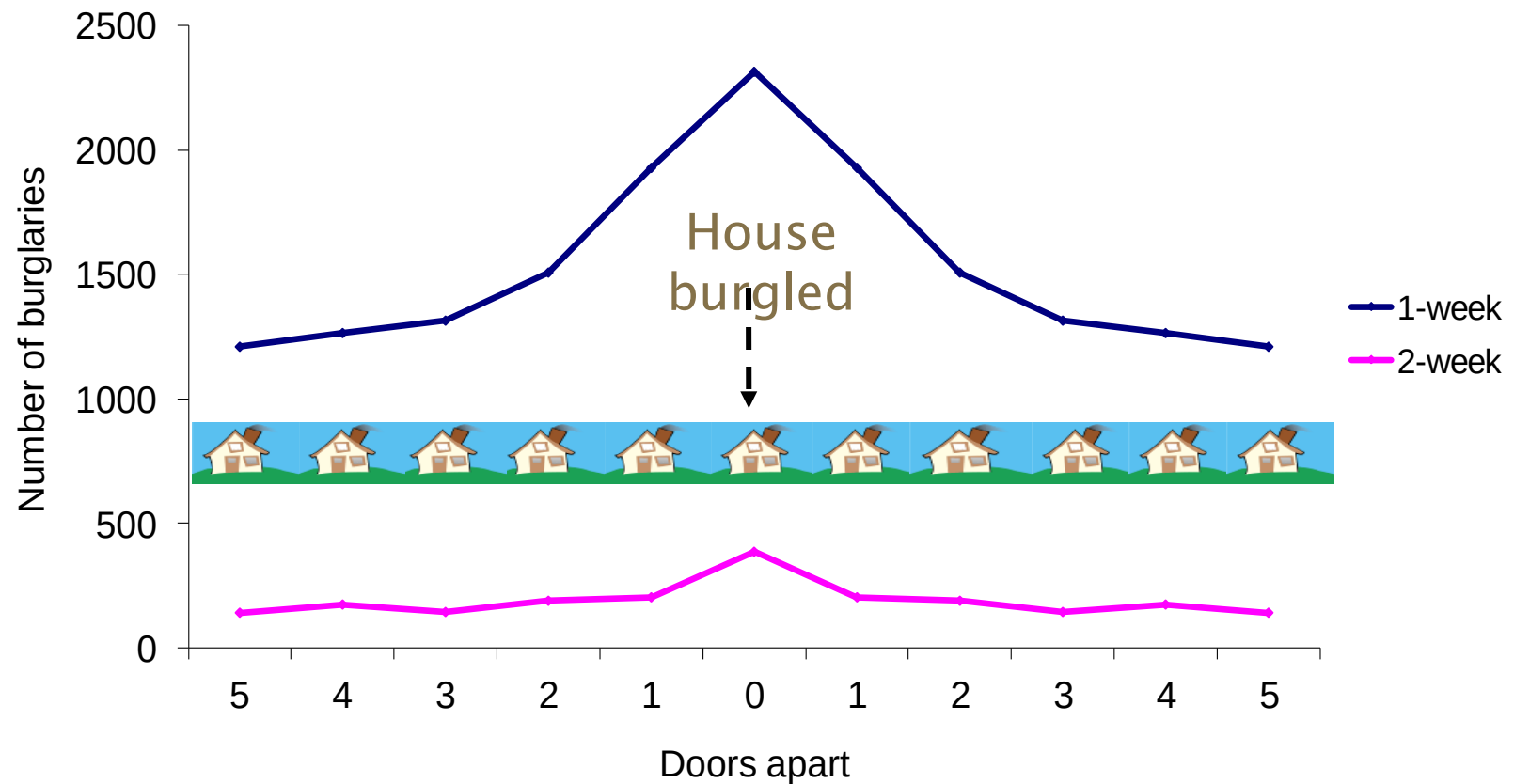
The time of repeat victimisation



Johnson, Bowers & Hirschfield, 1997

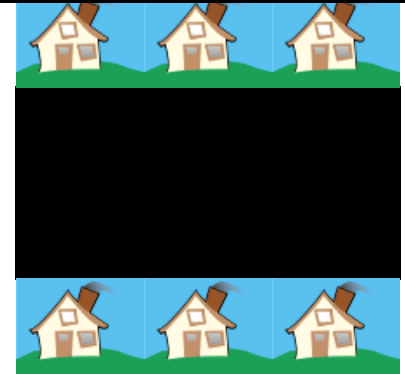
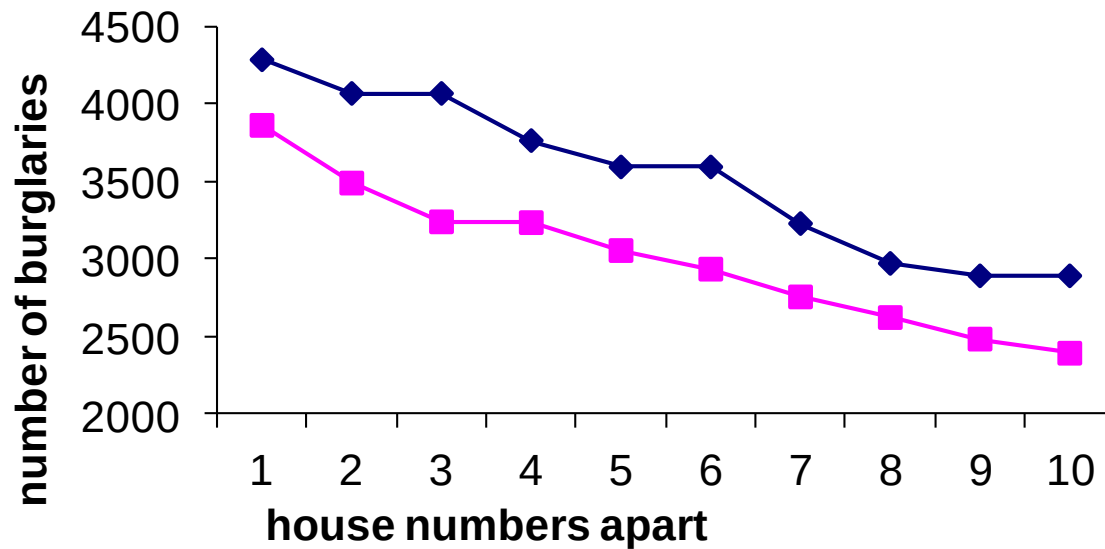
Results

The "neighborhood" effect



Results

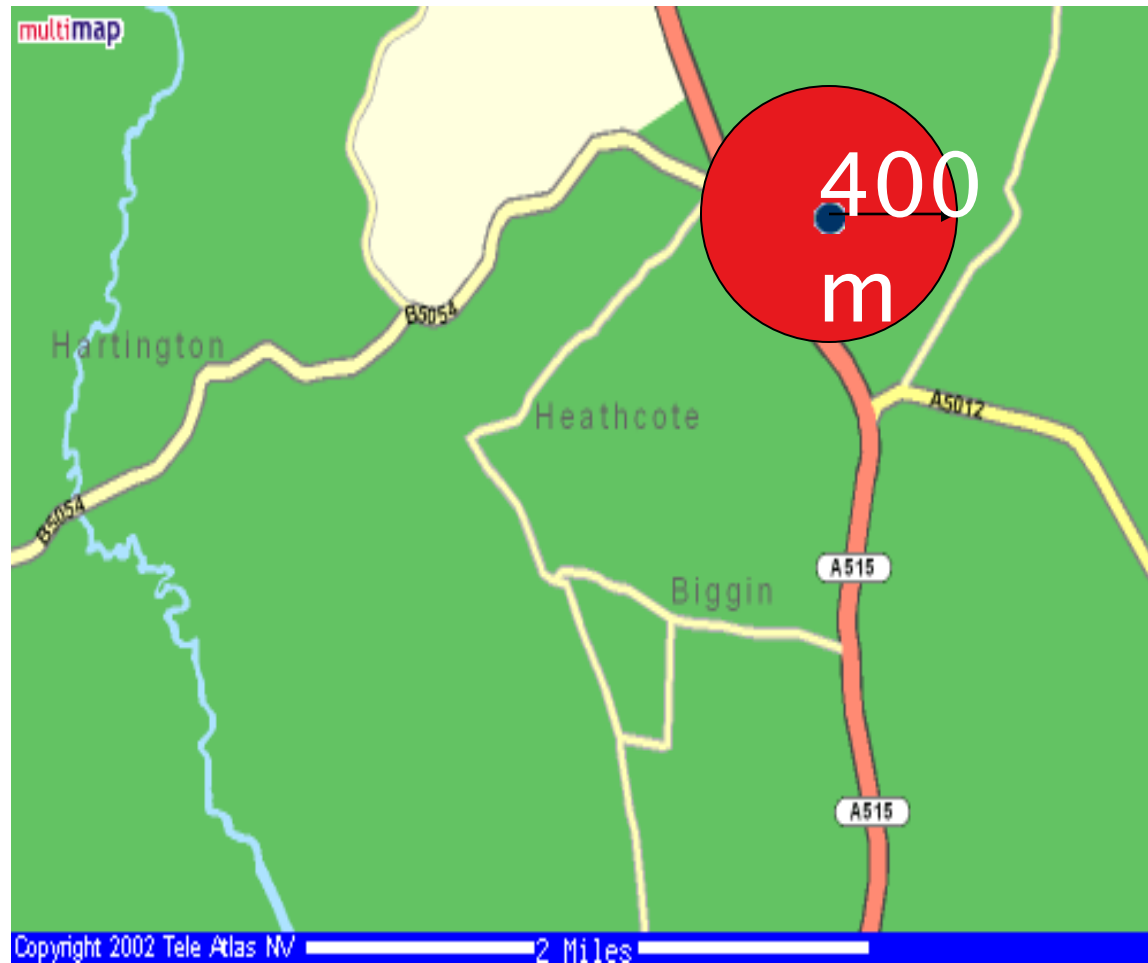
The “side of the road” effect



—◆— same side equivalent
—■— opposite side

Results

Area at risk



Results

- Burglary in a home is "transmissible," like a disease: houses within 400m of a victimized home are at increased risk for up to a month after the first robbery.
- The prospective-hotspot predicts 62% of robberies that occur in the following 2 days and 64% of those that occur in the following 7 days.

Focusing on repeat victimisation



Patterns of online repeat victimisation and implications for crime prevention

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6. From cyber-victimisation to its prevention: your human firewall

From cyber-victimisation to its prevention: your human firewall

- Watching and discussing *Your Human Firewall – The Answer to the Cyber Security Problem* | Rob May | TEDxWorking, available at <https://www.youtube.com/watch?v=BpdcVfq2dB8>